TILDE COMMUTES WITH TENSOR, TENSOR ALGEBRA AND INTERNAL HOM CONSTRUCTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1370-1384, following the statement at 1329-1368.

Editorial note. The lemma carries a second proof (source lines 1386 onward); per the same-proposition rule only the first proof is counted as one card.

Allowed standard prerequisites (editorial).

- Tilde as pullback along a morphism of ringed spaces. lemma-compare-constructions (same chapter, cited at ~1371) with Modules, Lemma modules-lemma-construct-quasi-coherent-sheaves (cross-chapter, cited at ~1372). Views M |-> tilde(M) as pi^* for pi: (X,O_X) -> ({*}, R).
- Pullback commutes with tensor and tensor algebra. Modules, Lemmas modules-lemma-tensor-product-pullback and modules-lemma-pullback-tensor-algebra (cross-chapter, cited at ~1376-1377). Gives items (1) and (2).
- Flatness of the structure sheaf over R. lemma-spec-sheaves (same chapter, cited at ~1380). Stalks are localizations, hence flat; needed for internal hom.
- Pullback commutes with internal hom for finitely presented first argument. Modules, Lemma modules-lemma-pullback-internal-hom (cross-chapter, cited at ~1383). Gives item (3).

Source reference. Original lemma 01I7: sources/stacks-schemes/pinned-original-source.tex, source lines 1295–1295. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01HV: sources/stacks-schemes/pinned-original-source.tex, source lines 909–909. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01BH: focused-seeds/geometry/S9-schemes-widetilde-constructions/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0C6I: focused-seeds/geometry/S9-schemes-widetilde-constructions/ref-tags.txt, source lines 6–6. Navigation pointer only: the

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01CI: focused-seeds/geometry/S9-schemes-widetilde-constructions/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01CD: focused-seeds/geometry/S9-schemes-widetilde-constructions/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01CM: focused-seeds/geometry/S9-schemes-widetilde-constructions/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01CF: focused-seeds/geometry/S9-schemes-widetilde-constructions/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01CA: focused-seeds/geometry/S9-schemes-widetilde-constructions/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(X, \mathcal{O}_X) = (\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ be an affine scheme. There are canonical isomorphisms*

1. *$\widetilde{M \otimes_R N} \cong \widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}$, see Modules, Section 01CA.*
2. *$\widetilde{T^n(M)} \cong T^n(\widetilde{M})$, $\widetilde{Sym^n(M)} \cong Sym^n(\widetilde{M})$, and $\widetilde{\wedge^n(M)} \cong \wedge^n(\widetilde{M})$, see Modules, Section 01CF.*
3. *if M is a finitely presented R-module, then $\mathcal{H}om_{\mathcal{O}_X}(\widetilde{M}, \widetilde{N}) \cong \widetilde{\mathrm{Hom}_R(M, N)}$, see Modules, Section 01CM.*

First proof. Using Lemma 01I7 and Modules, Lemma 01BH we see that the functor $M \mapsto \widetilde{M}$ can be viewed as π^* for a morphism π of ringed spaces. And pulling back modules commutes with tensor constructions by Modules, Lemmas 01CD and 01CI. The morphism $\pi : (X, \mathcal{O}_X) \to (\{*\}, R)$ is flat for example because the stalks of $\mathcal{O}_X$ are localizations of R (Lemma 01HV) and hence flat over R. Thus pullback by π commutes with internal hom if the first module is finitely presented by Modules, Lemma 0C6I. □